

1st Class Test -5th Sem
Design of Machine Elements- I

Time: 90 Mins

MM: 30

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| Q-1 | (i) | What are the various design considerations for mechanical design? | 5 |
| | (ii) | What are the factors influencing the selection FOS? | 5 |
| Q-2 | | In an eccentrically loaded riveted joint, A bracket is riveted to a column by 6 rivets of equal size symmetrically. It carries a load of 60 KN at a distance of 200 mm from centre of column. If the maximum shear stress in the rivet is limited to 150 MPa, determine the diameter of the rivet. Assume if any dimension required. | 10 |
| Q-3 | | Explain the design procedure of Longitudinal & Circumferential joints for pressure vessels. | 10 |